

SatQuery AI

Presentation Reference Document

"Ask your satellite imagery. SatQuery does the analysis."

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13 Slides · Problem → Solution → Features → USP → Workflow → Tech Stack → Feasibility → Risk → Comparison

The Core Issue We Are Solving

Satellite imagery is powerful, but difficult to interrogate.

Answering a seemingly simple question like:

"What changed between these two satellite images, where did it happen, and how significant was it?"

...can require several specialised tools, algorithms, and manual interpretation steps.

Why it's hard:

- Enormous volumes of optical, multispectral, and SAR data are generated daily
- Extracting insights still requires specialised GIS knowledge and multiple tools
- Manual analysis becomes slow and error-prone when comparing images or modalities
- Non-expert users cannot convert a natural question into the correct RS workflow
- Different tools are needed for visualisation, change detection, measurement, and interpretation

The Core Gap:

There is a gap between the **complexity of satellite data** and the **simplicity of the questions users want to ask**.

The Real Need — Beyond More Data

What Exists Today

- Vast volumes of satellite imagery being collected
- Powerful GIS and RS processing tools available
- Growing availability of open satellite data
- AI models that can describe images

What Is Still Missing

- Simple way for non-experts to extract insights
- Unified interface across modalities (optical + SAR)
- Trustworthy numerical answers — not hallucinations
- End-to-end workflow in a single conversational query

The Real Need

We don't just need **more satellite data**.

We need a **simpler way to interact with and understand that data**.

Make satellite-image analysis as intuitive as asking a question.

SatQuery AI — The Solution

"Ask your satellite imagery. SatQuery does the analysis."

SatQuery AI is an **interactive Vision-Language Assistant for remote sensing** that allows users to analyse satellite imagery through natural-language queries. Instead of manually selecting GIS operations, algorithms, and visualisation tools, the user simply asks.

The Four-Step Experience

ASK	ANALYZE	LOCATE	EXPLAIN
Type your question in plain language	SatQuery selects the right analysis to identify where changes or features are	Identify where changes or features are	Get the answer with evidence and context

Concrete Example:

User Query	What SatQuery Does
"What changed between these two images?"	Validates imagery → selects bi-temporal change analysis → computes pixel differences → calculates change_fraction → interprets evidence → presents answer with visual map

In one line: SatQuery converts a complex remote-sensing analysis workflow into a simple conversational interaction.

How SatQuery Addresses Each Need

Problem / Need	SatQuery Solution
Difficult satellite-image interaction	Natural-language query interface
Specialised GIS knowledge required	AI-driven automatic tool selection
Multi-step manual analysis pipeline	Automated end-to-end analysis workflow
Need for precise quantitative results	Deterministic geospatial computation (not AI-guessed)
Understanding temporal changes	Bi-temporal change analysis with pixel-level metrics
Multiple satellite modalities	Unified optical + multispectral + SAR support
Trustworthy and auditable AI answers	Evidence + provenance + 5-stage execution trace
No unified tool for interrogation	Single conversational platform across all analysis types

The result:

Users move from managing a complex technical workflow across multiple specialised tools to simply asking a question in plain language and receiving a complete, evidence-backed, scientifically grounded answer.

Five Core Capabilities

1

■ Natural-Language Satellite Querying

Users interact with satellite imagery using ordinary language instead of GIS commands. Queries are converted into appropriate analysis actions based on available imagery and the requested task.

2

■ Deterministic Geospatial Analysis

Quantitative results — changed pixels, change fraction, area, spectral indices, SAR statistics — are calculated using deterministic geospatial tools, never generated by a language model.

3

■ Multi-Image Bi-Temporal Change Detection

SatQuery compares bi-temporal satellite imagery to identify, quantify, and map spatial changes. Produces pixel counts, change fractions, and visual change evidence maps.

4

■ Optical + SAR Multimodal Analysis

Works with optical/multispectral and SAR imagery simultaneously, allowing observations from different sensing modalities to be compared as complementary spatial evidence.

5

■ Evidence-Driven AI Interpretation

The Vision-Language Model interprets computed measurements and visual evidence to produce a natural-language answer — with full provenance and an execution trace showing every step.

What Makes SatQuery Different

■+■ AI + Geospatial Tools — Not AI Alone

AI understands the question and the imagery. Specialist geospatial tools perform the actual spatial analysis and measurements. SatQuery is not a VLM looking at an image — it is a VLM coordinating with precision geospatial computation.

■≠■ Measurement ≠ Hallucination

Computation happens first. Evidence is built. AI then interprets and explains those results. We don't ask AI to guess the measurement — we ask AI to explain the measurement that was already computed deterministically.

■■■ No GIS Expertise Required

Users don't need to know GIS commands, spectral indices, SAR processing, or image registration. They simply ask: 'What changed?' or 'Compare these two images.' SatQuery handles the rest.

■ Ask Once. Get the Whole Insight.

Traditional workflow: Open imagery → select tool → process → calculate → visualise → interpret → report. SatQuery: Ask → Analyze → Locate → Explain. One query returns the answer, measurements, spatial evidence, and interpretation together.

**"We are not making satellite analysis more powerful for experts.
We are making it simpler for everyone."**

From User Query to Actionable Insight

Traditional Workflow

1. Open GIS software
2. Select algorithm
3. Configure parameters
4. Run analysis
5. Visualise results
6. Interpret manually
7. Write report

Multiple tools · Expert knowledge · Hours

SatQuery Workflow

1. Upload satellite image(s)
2. Ask a question in plain language
3. SatQuery understands intent
4. Selects appropriate analysis
5. Computes evidence (deterministic)
6. AI interprets the evidence
7. Answer + evidence + visual output

One interface · No expertise needed · Seconds

SatQuery Step-by-Step

USER INPUT Upload satellite image(s) + Ask a natural-language question

UNDERSTAND Identify what the user wants and what imagery is available

SELECT Choose the right analysis: VQA / Change Detection / SAR / Cross-Modal

ANALYZE Specialist tools process imagery and compute evidence deterministically

INTERPRET Vision-Language Model interprets computed evidence and imagery

PRESENT Answer + Location + Measurements + Visual Evidence + Execution Trace

Technology Stack

■ FRONTEND

- Next.js 16 (App Router)
- React + TypeScript
- Responsive Map Dashboard
- Image Canvas + Query Bar
- Evidence & Trace Viewer

■ BACKEND

- Python 3.11+
- FastAPI (REST API)
- Uvicorn (ASGI Server)
- Pydantic v2 Models
- Modular Agent Interfaces

■ AI / VLM

- Qwen2-VL-2B-Instruct
- PEFT LoRA ($r=8$, $\alpha=16$)
- RS Domain Adapter
- HuggingFace Transformers
- CPU + GPU Inference

■ GEOSPATIAL

- Rasterio — GeoTIFF I/O
- Shapely — Geometry / IoU
- NumPy — Numerical Arrays
- SciPy — Connected Regions
- CRS & Affine Transform

■ AGENT / TOOLS

- Heuristic Modality Router
- Tool Registry
- Deterministic Specialists
- Evidence Generation
- 5-Stage Execution Trace

■ TESTING

- pytest (54 pass, 0 fail)
- Unit + Integration + Regression
- VRSBench / RSVQA / CDVQA
- Frontend Build Validation
- verify_demo.bat ALL [PASS]

■ No mandatory external API key required · Fully offline capable · Open-source core architecture

Feasibility of SatQuery AI

HIGH

■ Technical Feasibility

- Core AI, geospatial, multimodal, and web components are fully implemented and integrated
- Natural-language VQA, change detection, SAR analysis, and optical+SAR fusion — all working
- 54 tests passing, 4 demo flows verified end-to-end against real satellite data
- Frontend builds with 0 errors; backend health check confirmed
- `verify_demo.bat`: ALL [PASS]

HIGH*

■ Operational Feasibility

- Runs locally on a standard developer laptop — no cloud infrastructure required for demo
- Windows and macOS supported; no mandatory external API key
- Complete offline capability for the SIH demonstration environment
- * Production-scale deployment requires: GPU workers, async job queuing, persistent storage
- * Multi-user management and production monitoring would be needed for operational rollout

HIGH***■ Legal Feasibility**

- Primarily open-source technologies (FastAPI, Next.js, Rasterio, NumPy, Shapely)
- Locally executed VLM — user imagery is not sent to any external API by default
- * Formal licence verification required for: UCM-Captions dataset (training data)
- * Model licence (Qwen2-VL) and all third-party library terms require review before commercialisation
- * No hidden ISRO/SAC data claims — benchmark results reported honestly as NOT RUN

■ FEASIBLE NOWwith documented scaling and licensing considerations for
production

Potential Challenges & Risks

We acknowledge these challenges honestly — and have identified concrete strategies to address each one.

■ Inference Scalability & Latency

VLM inference on CPU takes 60–90 seconds. At scale or in real-time applications, this becomes a bottleneck that limits usability.

■ Strategy

Deploy GPU-accelerated inference workers, implement asynchronous job queuing, apply model quantisation and result caching for repeated queries.

■ Sensor & Data Diversity

Different satellites, resolutions, sensors, projections, and acquisition conditions can produce inconsistent or unreliable analysis results.

■ Strategy

Build sensor-aware preprocessing pipelines, extend metadata validation, develop modality-specific specialist tools, and enforce standardised spatial alignment.

■ Real-World Image Variability

Clouds, shadows, seasonal differences, terrain displacement, noise, and co-registration errors can appear as false positives in change detection.

■ Strategy

Introduce cloud/quality masking, improved co-registration algorithms, terrain-aware processing layers, and uncertainty-aware thresholding in change detection.

■ AI Reliability & Uncertainty

The VLM interpretation and heuristic confidence values cannot guarantee perfect semantic understanding across all query types and imagery conditions.

■ Strategy

Maintain strict separation of deterministic measurements from AI interpretation, strengthen evidence grounding, and implement calibrated confidence estimation.

■ Generalisation & Benchmark Validation

Performance must be systematically validated on diverse real-world datasets including VRSBench, RSVQA, CDVQA, and hidden ISRO/SAC imagery.

■ Strategy

Run formal evaluation on all three benchmarks once datasets are available. Prioritise real ISRO/SAC hidden-data validation as the primary production target.

"We don't solve uncertainty by making the AI more confident. We solve it by making the evidence stronger."

What Existed vs What SatQuery Adds

Capability	Conventional GIS / RS	AI / VLM Only	SatQuery AI
Natural-language interaction	■ None	■ Yes	■ Yes
Ask questions about satellite imagery	■ Manual + expert	■ Yes	■ Yes
Automatic workflow selection	■ Manual	■ Model-dependent	■ Yes
Deterministic geospatial measurements	■ Yes	■ Not central	■ Yes
AI-based visual interpretation	■ None	■ Yes	■ Yes
Bi-temporal change analysis	■ Yes	■ Research-specific	■ Yes
Optical imagery support	■ Yes	■ Yes	■ Yes
SAR imagery support	■ Specialised	■ Limited	■ Yes
Optical + SAR cross-modal fusion	■ Manual workflows	■ Very limited	■ Integrated
Evidence-backed AI response	■ Tool outputs	■ Varies	■ Yes
Numerical provenance / traceability	■ Tool-dependent	■ Limited	■ Full trace
Visual evidence with answer	■ Separate step	■ / Varies	■ Integrated
Multi-step workflow in one query	■ Multiple tools	■ Partial	■ Yes
GIS expertise required from user	HIGH ■	LOW ■	LOW ■

From Tool-Centric to Question-Centric Analysis

The key shift SatQuery enables:

	Traditional RS / GIS	AI-Only Approach	SatQuery AI
Starting point	Satellite image files	Image + question	Image(s) + plain question
User action	Open GIS → choose algorithm	Send image to AI model	Type a question
Analysis step	Run tool(s) manually	AI generates answer	Automatic: select → compute → interpret
Numerical results	Tool output (if applicable)	AI-generated (may hallucinate)	Deterministic computation
Explanation	Expert interprets manually	AI explanation	AI explains verified evidence
Output	Data file / report	Text description	Answer + location + metrics + visual + trace
Expertise required	High — GIS specialist	Low — but trust limited	None — anyone can ask
Time to answer	Minutes to hours	Seconds (but less reliable)	Seconds to minutes (fully grounded)

The headline for your presentation:

From Tool-Centric Analysis → to Question-Centric Analysis

Recommended Slide Sequence & Key Messages

Slide	Title	Key Message
1	The Problem	Satellite data is powerful but hard to interrogate
2	Why Solve It	More data is not enough — we need simpler access
3	Our Solution	Ask your satellite imagery — SatQuery does the analysis
4	Problem → Solution Map	Every need maps to a specific SatQuery capability
5	Key Features	5 pillars: NL query · Deterministic · Change · SAR · Evidence
6	Unique Value (USP)	AI + Geo Tools · No hallucination · No expertise · One query
7	End-to-End Workflow	User Input → Understand → Select → Analyze → Interpret → Present
8	Technology Stack	Next.js · FastAPI · Qwen2-VL · LoRA · Rasterio · NumPy · SciPy
9	Feasibility	Technical HIGH · Operational HIGH* · Legal HIGH*
10	Challenges & Mitigation	5 risks identified with 5 concrete strategies
11	Competitive Comparison	Existing tools OR AI — SatQuery combines both + adds provenance
12	Evolution Story	Tool-centric → Question-centric analysis
13	This summary	Narrative guide for judges and presenters

Judge Narrative — One Paragraph:

"SatQuery AI bridges the gap between expert remote-sensing tools and accessible AI interfaces. The VLM understands imagery and natural-language intent, while deterministic geospatial tools calculate the numbers. Every answer is accompanied by the evidence and execution trace that produced it — making satellite image analysis as simple as asking a question, while keeping the science honest and verifiable."